

RAP 320 Remote Support & Remote Software Modifications

From Rovipedia, the Rovisys encyclopedia

Approved

Purpose

This document establishes good engineering practices when making system modifications from a remote location.

Quick Links

[Remote Modifications](#)

[Initiating Remote Support](#)

[Logging Into The Customer's System](#)

[Online Support](#)

[Document The Modifications](#)

Definitions:

Good Engineering Practice (GEP): The abbreviation GEP will be used throughout this document to highlight the steps to be used when performing remote support.

Remote Modifications

Remote modifications may be performed under multiple circumstances the most common being:

1. As part of remote startup and commissioning.

For remote startup and commissioning support, all remote activities should be carried out according to the project's commissioning plan(s) and customer agreed workflows. The remote commissioning plan may use this article or other customer policies as a basis when being created.

2. In relation to a customer support request.

For remote support, all remote activities should be carried out in accordance with the best practices as outlined in this article.

Initiating Remote Support

Remote support can be initiated in multiple ways, most commonly:

1. The customer makes initial contact, requesting support for some issue with the system, this could be scheduled or unscheduled.
2. A RoviSys engineer contacts the customer with an update or resolution to an issue for the customer system.

Regardless of the method of initiating the remote support, all remote support actions must be approved by the customer.

GEP #1

New support requests should follow RAP314 for logging and routing to the supporting team member(s).

GEP #2

If the support request is non-emergency & not time-sensitive in nature, a summary of changes should be prepared and agreed upon with the customer which includes an outline of what will be changed and what the potential operational impacts related to the changes might be. Testing requirements should also be documented and agreed upon ahead of time.

GEP #3

After the initial request for remote support has been made and action plan agreed with the customer, the RoviSys engineer must check in with the control room operators, plant controls engineer, IT, or other system owner to confirm that changes can be made. The approver(s) should be identified during the project or initialization of a support contract.

It is important to recognize that sometimes local plant production conditions may prohibit remote support and that the requesting party was unaware (if they are remote to the plant). When making changes that could affect production status (could include PLC, HMI, MES, etc.), notification and approval from local plant operations (control room / manufacturing supervisor, etc.) should be contacted for request and notification of the change to be made. In many cases the customer approver will handle this additional step, but it should not be assumed.

After clearing the proposed actions for remote support with the customer and the operations where applicable, it is safe to log into the customer's computer system.

Record the time when you connected to their system and who approved start of support work.

Logging Into The Customer's System

The actual method for logging into the customer system will vary for each customer and each site.

Most systems are accessible through a customer company network. This approach usually involves creating a virtual private network (VPN) using software supplied by the customer. The customer's IT department will create an account that allows access to the control system. In keeping with best industrial security practices, the use of multi-factor authentication, jump hosts or other defense in depth may be required and should be expected. RoviSys should typically not share logins among employees nor with any third parties.

Other less common options may require having a local resource use laptop with hotspot access and screen-sharing software such as Zoom or TeamViewer.

GEP #4

All of the necessary information for establishing a connection to the customer's system should be stored in GrandView or another approved RoviSys system. Include documents with special instructions and/or software where necessary. (VPN software used, MFA instructions if applicable, list of users with remote access, project PC/asset usernames and passwords, etc.)

GEP #5

Prior to making any changes on the remote system, create a backup.

This may include running backup utilities or copying files into a backup directory. Include any component that might change such as databases, PLC code, graphics, custom config files (e.g. .ini, .xml, .cfg), etc. While making system backups, invoke a revision methodology to track changes in conjunction with any customer change control policies:

examples: (or follow revision method as defined in the Project Plan if support is continuation of a previously project

- Folder names that use the date as a title; yyyyymmdd (IEC date format)
- Project names that use a revision number; RVS22A_rev2

Online Support

When working online with a remote system, it is important to take extra precautions: Take notice of the process condition and determine if the changes to be implemented will upset the process.

In the event a process upset occurs because of the changes, immediately notify the customer and control room to initiate a controlled response.

GEP #6

Follow any established protocols provided by the customer or project manager.

Monitor for new alarms and respond according to plant procedure. (Notify operations, acknowledge if permitted, etc.)

Only operate equipment if explicitly permitted and in coordination with the plant.

GEP #7

As one would expect, working on a system remotely is slower than being in front of the actual machine. Be patient while making changes; avoid clicking multiple times to “speed up” results.

GEP #8

All changes need to be tested or validated before ending the remote support session. An approved test plan should be part of the pre-login discussions with the customer project manager or other authorized customer employee. If remote support is related to troubleshooting, confirming the return to standard operation may be adequate, if agreed by the customer.

When the modifications are complete and operation is verified, return the system to the appropriate operational state. This may include stopping an application that is running in demo mode to allow changes and restarting it in runtime.

Do not make any additional or new modifications to any other process controls areas other than the one(s) already documented and approved by the client while you are online. Use common sense, if documenting and approval is possible during the support, it can be completed.

GEP #9

Notify the control room operators and/or customer controls contact (same contact as GEP #3) that the changes are complete, and you are logging off the system. Leave them with a contact number in the event something occurs after you log off. Record the time when you disconnected from their system and who you notified.

Document The Modifications

It is critical that any changes to a remote system be documented. This documentation is used to convey the change to the customer as well as provide a permanent record in the project history for future reference by RoviSys employees.

GEP #10

Changes to remote systems must be documented using a GrandView status log, the On-Site Report template report a Project Support logging Excel sheet, or similar document including testing results. Approved internal RoviSys software support tracking tools may also be used to log and document the tickets and results.

A copy of the report or log should be sent to the customer or accessible by the customer from the tracking system.

The report should contain the following items:

- Record the start and end date and time of system access, as well as the name(s) of the plant personnel that you confirmed approval to proceed. Remember: The person who requested the change may be at a different location from the plant you are supporting.
- Summary of items changed. Include enough detail to fully define what was performed or reference the predefined scope if applicable. Note any special procedures required such as using demo mode or restarting applications and/or servers.
- Summary of tested results and any deviations from the initial plan.
- Any follow up items required (or new discoveries for another support request).

Page History

Revision	Action/Comment	Date	User
r15 (this)	Approve	1-Mar-2023 7:53 UTC	Joshua Hilewick
r15 (this)	Create/Edit <i>PECH-Minor edits</i>	28-Feb-2023 5:47 UTC	Dave Hermann
r14	Approve	25-Jan-2023 8:56 UTC	Beverly Kiglics
r14	Create/Edit <i>fixing quick links</i>	28-Dec-2022 4:54 UTC	Joshua Hilewick
r13	Create/Edit <i>overhaul of document, quick links added</i>	28-Dec-2022 4:53 UTC	Joshua Hilewick
r12	Create/Edit <i>complete overhaul for current best pr...</i>	28-Dec-2022 4:23 UTC	Joshua Hilewick
r11	Create/Edit <i>PECH-Periodic Review</i>	15-Nov-2022 5:31 UTC	Dave Hermann
r10	Approve	16-Jan-2019 11:05 UTC	Jeff Marshall
r10	Create/Edit <i>Formatting changes</i>	10-Jan-2019 6:00 UTC	Devon Graves
r9	Create/Edit <i>Removed HTML</i>	8-Jan-2019 4:33 UTC	Malec Kennedy
r8	Create/Edit <i>Formatting changes</i>	29-Nov-2018 2:31 UTC	Jared Simko
r7	Approve	7-Jul-2016 4:03 UTC	Jeff Marshall
r7	Create/Edit <i>uncapitalize</i>	15-Jun-2016 6:10 UTC	Malec Kennedy
r6	Approve	16-Jan-2012 8:23 UTC	Pete Madonia
r5	Approve	16-Jan-2012 8:22 UTC	Pete Madonia
r6	Create/Edit <i>typos</i>	9-Jan-2012 8:58 UTC	Alexandria Caliguire
r5	Create/Edit <i>grammer fix</i>	20-Dec-2011 8:18 UTC	Tim Mirth
r4	Approve	7-Mar-2011 2:20 UTC	John Turk
r4	Create/Edit <i>Import rev. c of RAP into RoviPedia</i>	2-Sep-2010 7:40 UTC	Tom Grulkowski
r3	Create/Edit <i>updated tags</i>	27-Aug-2010 2:11 UTC	Haren, Michael
r2	Create/Edit <i>updated tags</i>	27-Aug-2010 3:51 UTC	Haren, Michael
r1	Create/Edit <i>Initial revision</i>	26-Aug-2010 5:33 UTC	Xander Dumaine

Note: This is a snapshot of a living document. Please check online for the latest revision.

Edited: 28-Feb-2023 5:47 UTC **Printed:** 20-Oct-2023 6:54 UTC

Document Number: 124 Revision: r15